Case Study:

Designing a Customer Intelligence Pipeline for an E-Commerce Company

Scenario:

An e-commerce company wants to improve customer onboarding and provide personalized product recommendations. The company collects data from multiple sources, including customer registration forms, website clickstreams, payment transactions, customer reviews, and inventory systems.

The management wants a data-driven solution that can collect, process, store, and analyze customer information to improve customer experience and support business decision-making.

Your Task

Using the concepts learned in Data Engineering Fundamentals, design an end-to-end data engineering solution for the company.

1. Data Sources

- Identify all possible data sources involved in this scenario.
- Categorize each source as **Structured**, **Semi-Structured**, or **Unstructured**.

Data Source	**Data Type**
Customer Registration Forms	Structured
Payment Transactions	Structured
Inventory Database	Structured
Website Clickstreams	Semi-Structured
Mobile App Activity	Semi-Structured
Customer Reviews	Unstructured

2. Data Ingestion

- Which ingestion method would you use for each source:
    - Batch Processing
    - Stream Processing
    - File-Based Ingestion
- Justify your choices.

Data Source	Ingestion Method	Reason
Registration Forms	Stream Processing	New customer data arrives continuously
Website Clicks	Stream Processing	Requires real-time tracking
Payment Transactions	Stream Processing	Immediate processing needed
Inventory Reports	Batch Processing	Updated periodically
Customer Reviews	File-Based / Batch	Collected and loaded periodically

3. ETL Process

- What data would be extracted from each source?
- What transformation steps would be required?
- Where would the transformed data be loaded?

Extract

Collect data from:

- CRM Systems
- Website Logs
- Payment Gateway
- Inventory Database
- Customer Reviews

Transform

- Remove duplicate customer records

- Validate email addresses
- Standardize phone numbers
- Clean incomplete records
- Categorize products and customer segments

Load

Store processed data in:

- Data Lake (raw data)
- Data Warehouse (analytics-ready data)

4. Data Storage

- Which data should be stored in a Database?
- Which data should be stored in a Data Lake?
- Which data should be stored in a Data Warehouse?

  Database

  - **Customer Profiles**
  - **Order Information**
  - **Product Details**

  Data Lake

  - **Website Clickstream Logs**
  - **Customer Reviews**
  - **Product Images**
  - **Mobile App Data**

  Data Warehouse

  - **Sales Reports**
  - **Customer Analytics**
  - **Product Performance Reports**
  - **Revenue Dashboards**

5. Data Lake

- What types of raw data would be stored in the Data Lake?
- Which Data Lake zones (Raw, Cleansed, Curated, Sandbox) would be used during data processing?

Raw Data Stored

- Website Clicks
- Customer Reviews
- Product Images
- Transaction Logs
- Sensor Data

Zone	Purpose
Raw Zone	Store data as received
Cleansed Zone	Clean and validate data
Curated Zone	Analytics-ready data
Sandbox Zone	Data science experiments

6. Data Warehouse

- What business reports could be generated from the Data Warehouse?
- Which departments would benefit from the warehouse data?

Business Reports Generated

- Monthly Sales Reports
- Customer Segmentation Analysis
- Product Performance Dashboard
- Revenue Trends
- Regional Sales Reports

Departments Benefiting

- Sales Team

- Marketing Team
- Finance Team
- Operations Team
- Management Team

7. Data Modeling and Data Pipelines

- Design a simple data model for Customers and Orders.
- Draw an end-to-end data pipeline showing how data moves from source systems to analytics.

Customer Table : Customer ID , Name, Email, Country
Orders Table : Order ID, Customer ID, Product, Amount
(one to many relationships)

1. **Data Source:** Registration Form, Website Clicks, Payment System, Customer Reviews
2. **Data Ingestion**
3. **ETL**
4. **Data Lake**
5. **Data Warehouse**
6. **Dashboard**
7. **AI Model**

8. Data Quality and Validation

- Define at least five validation rules for customer data.
- Identify possible data quality issues and how they can be resolved.

Validation Rules

1. Customer ID must be unique.
2. Email must contain '@'.
3. Phone number must be 10 digits.
4. Mandatory fields cannot be empty.
5. Transaction amount cannot be negative.

Possible Data Quality Issues

Issue	Solution
Missing Values	Data Validation
Duplicate Records	Deduplication
Incorrect Formats	Data Standardization
Invalid Emails	Validation Rules
Inconsistent Data	Data Cleansing

9. Analytics and AI Applications

- What analytics dashboards can be built using this data?
- Suggest at least three AI applications that can use the processed data.

Analytics Dashboards

- Customer Acquisition Dashboard
- Sales Performance Dashboard
- Product Recommendation Dashboard
- Revenue Analysis Dashboard
- Customer Retention Dashboard

AI Applications

Product Recommendation System

Suggest products based on customer browsing and purchase history.

Customer Segmentation

Group customers based on behavior and purchasing patterns.

Customer Churn Prediction

Predict customers who are likely to stop purchasing.

Fraud Detection

Identify unusual transactions and suspicious activities.